

Pythagoras' theorem



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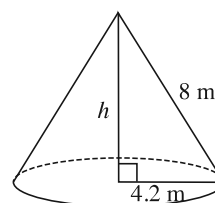
UNIT 10: Applications of Pythagoras' theorem

QUESTION 1 Find the length of the diagonal (to 1 decimal place) of:

a a square of side length 5 cm

b a rectangle 28 cm long and 9 cm wide

QUESTION 2 The radius of the base of a cone is 4.2 m and its slant height is 8 m. Find the height of the cone correct to one decimal place.



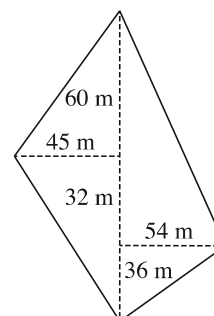
QUESTION 3 Find the length of the third side of a right-angled triangle, (to one decimal place), if the:

a longest side is 49 cm, other side is 12 cm

b hypotenuse is 45 cm, other side is 16 cm

QUESTION 4 What is the altitude of an equilateral triangle where sides are each 12 cm (answer correct to 2 decimal places).

QUESTION 5 Find the perimeter of this block of land.



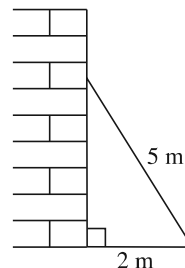
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UNIT 11: Problem solving

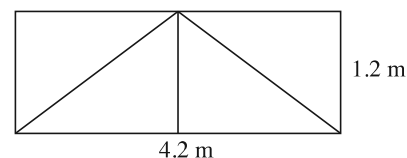
QUESTION 1 A 5 metre ladder has its foot 2 metres from the foot of a wall. How far up the wall does the ladder reach (give the answer to the nearest cm)?



QUESTION 2 Two roads are at right angles to each other. Person A walks 8 km on one road and person B walks 15 km on the other road. How far apart are A and B?

QUESTION 3 A 6 metre ladder rests against a wall and its foot is 3 metres away from the base of the wall. How high does the ladder reach the wall (answer correct to 2 decimal places)?

QUESTION 4 Carlo is building a rectangular gate from steel pipe. The gate is 4.2 m long and 1.2 m high. In order to brace the gate, Carlo wants to add a centre brace and two diagonal braces as shown in the diagram. He has 6 m of pipe left. Is this enough for the bracing he wants to do? Justify your answer.



QUESTION 5 Two flag posts are 9 m and 12.5 m long and 24 m apart. Find the length of the string needed to join the tops of the two posts.

